

PRIMEVIZ

Full Project Documentation

Project Type: End-to-End Data Analytics

Domain: E-Commerce / Retail Sales

Period Covered: February – April 2025

Tools: MySQL 8.0+ | Power BI Desktop | Advanced SQL

Records Analyzed: 250 Orders | Total Revenue: \$243,845

Turning raw sales data into actionable intelligence

Portfolio Project Document | 2025

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1. Executive Summary

This document presents a comprehensive end-to-end data analytics project focused on Amazon-style e-commerce sales data for the first quarter of 2025 (February through April). The project, named **PrimeViz**, combines rigorous SQL-based analysis with interactive Power BI visualization to extract actionable business intelligence from 250 transactional records representing \$243,845 in total revenue.

The analysis addresses ten carefully designed business questions covering revenue performance, product and category contribution, customer behavior, geographic distribution, payment method effectiveness, and order status health. Advanced SQL techniques including Common Table Expressions (CTEs), window functions (DENSE_RANK), conditional aggregation, and percentage-of-total calculations were applied to produce precise, decision-ready results.

Key findings reveal that Electronics and Home Appliances together account for more than 96 percent of total revenue, while a cancellation rate of 30.8 percent represents a significant operational risk. Completion rate stands at only 35.2 percent, indicating substantial room for process improvement. Top-performing products include Refrigerator, Laptop, and Smartphone. High-value customers such as Olivia Wilson, Jane Smith, and Emma Clark drive disproportionate revenue contribution.

The project delivers a four-page interactive Power BI dashboard (Overview, Sales Analysis, Customer Insights, and Operations), a complete MySQL analysis script, structured documentation, and a prioritized set of business recommendations. The work demonstrates professional-grade database design, advanced SQL, business analysis, dashboard design, and data storytelling skills suitable for a data analyst portfolio.

Core Performance Snapshot

Metric	Value	Interpretation
Total Revenue	\$243,845	Primary success metric for the period
Total Orders	250	Complete transactional volume analyzed
Completion Rate	35.2%	Only 88 orders fully completed
Cancellation Rate	30.8%	Critical operational risk flag
Electronics Share	~53%	Dominant revenue category
Avg Order Value	~\$975	Relatively high AOV driven by appliances

Table 1: Core performance metrics for Q1 2025

2. Project Introduction

E-commerce businesses generate large volumes of transactional data that, when properly analyzed, can reveal patterns in customer behavior, product demand, operational bottlenecks, and revenue concentration. PrimeViz was designed as a realistic portfolio-grade project that mirrors the analytical workflow used by professional data analysts in retail and e-commerce environments.

The dataset simulates Amazon-style order records containing product details, pricing, quantities, customer information, payment methods, and order status. Although compact (250 rows), the data is rich enough to support meaningful aggregation, ranking, trend analysis, and geographic evaluation. The limited size allows full transparency of results while still requiring advanced SQL techniques and careful dashboard design.

The project deliberately combines two complementary skill areas: (1) rigorous SQL analysis performed inside a MySQL database, and (2) interactive visualization and storytelling delivered through Power BI. This dual approach reflects modern analytics practice where data is cleaned and transformed in a database or warehouse layer and then consumed by business intelligence tools for decision support.

Why This Project Matters

Many portfolio projects stop at basic descriptive statistics or simple charts. PrimeViz goes further by framing ten explicit business questions, solving them with professional SQL (including window functions), interpreting the results in a business context, and packaging the findings into a multi-page dashboard and formal documentation. The result is a coherent narrative from raw CSV to executive-ready insights.

The documentation that follows is structured so that a hiring manager, peer reviewer, or future self can understand the full analytical journey without needing to reverse-engineer code or dashboard screens. Every major decision—schema design, metric definition, ranking method, and recommendation priority—is explained and justified.

3. Business Objectives

The primary objective of PrimeViz is to transform transactional sales data into clear, prioritized business insights that support revenue growth and operational improvement. Specific objectives are listed below.

- **Revenue Understanding:** Quantify total revenue, average order value, and the contribution of each product category and individual product to overall sales.
- **Performance Ranking:** Identify the top products by quantity sold and the top customers by total spend so that marketing and inventory efforts can be focused.
- **Operational Health:** Measure completion, pending, and cancellation rates and surface any category-level patterns that explain status outcomes.
- **Payment Channel Effectiveness:** Evaluate which payment methods generate the highest revenue share and order volume.
- **Geographic Insight:** Rank cities by sales performance and average order value to guide regional prioritization.
- **Temporal Pattern Recognition:** Analyze monthly sales and order volume across February, March, and April 2025 to detect trends or seasonality signals.
- **Advanced Ranking:** Produce a category-level ranking of products using window functions so that the strongest items within each category are clearly visible.
- **Visualization & Communication:** Translate all findings into an interactive four-page Power BI dashboard and a professional written report suitable for stakeholder review.

Secondary objectives include demonstrating clean database design practices, consistent metric definitions, and professional documentation standards that would be expected in a real analytics team environment.

4. Project Scope & Boundaries

In Scope

The project covers the complete analytical lifecycle for the provided sales dataset: schema design and table creation in MySQL, data loading and quality checks, exploratory analysis, ten structured business questions solved with SQL, interpretation of results, design and construction of a four-page Power BI dashboard, formulation of business recommendations, and production of supporting documentation (README, insights document, technical notes, presentation, and this full report).

All analysis is confined to the period February 1, 2025 through April 2, 2025. Categories analyzed are Electronics, Home Appliances, Clothing, Footwear, and Books. Order statuses considered are Completed, Pending, and Cancelled. Geographic analysis is performed at the city level using the CustomerLocation field.

Out of Scope

- Predictive modeling or machine learning (e.g., cancellation prediction, demand forecasting).
- Customer segmentation techniques such as RFM or clustering.
- Real-time data pipelines or automated ETL scheduling.
- Integration with live Amazon APIs or external market data.
- A/B testing design or causal inference analysis.
- Mobile or web application development beyond the Power BI dashboard.
- Cost, margin, or profit analysis (cost data is not available).
- Inventory level or supply-chain optimization modeling.

These boundaries keep the project focused, fully reproducible, and appropriate for a portfolio demonstration of SQL and Power BI skills while remaining realistic in scope.

5. Dataset Description

The source dataset is a cleaned CSV file containing 250 order records. Each row represents a single order with product, customer, payment, and status attributes. The data is structured as a flat fact table suitable for direct loading into a relational database and subsequent aggregation.

Dataset Characteristics

Attribute	Detail
File Name	amazon_sales_2025.csv
Number of Records	250
Number of Columns	11
Time Range	2025-02-01 to 2025-04-02
Total Revenue	\$243,845
Total Quantity Sold	714 units
Distinct Products	10
Distinct Categories	5
Distinct Customers	Approximately 12 named customers with repeated purchases
Distinct Cities	Multiple major US cities
Payment Methods	PayPal, Credit Card, Debit Card, Amazon Pay, Gift Card
Order Statuses	Completed, Pending, Cancelled

Table 2: High-level dataset characteristics

The dataset is intentionally compact so that every query result can be manually verified and so that the focus remains on analytical technique rather than data engineering scale. Despite its size, the data exhibits realistic concentration patterns (heavy revenue in two categories, high cancellation rate, repeat customers) that make the business findings meaningful.

6. Data Dictionary

The following dictionary defines every column in the `amazon_sales` table after it has been loaded into MySQL. Data types were chosen deliberately: `DECIMAL` for monetary values to preserve precision, `DATE` for proper temporal filtering, and `VARCHAR` with reasonable lengths for text fields.

Column	Type	Description
OrderID	VARCHAR(20) PK	Unique order identifier (e.g., ORD0001)
Date	DATE	Order placement date (YYYY-MM-DD)
Product	VARCHAR(100)	Product name (Laptop, Smartphone, etc.)
Category	VARCHAR(100)	Product category (Electronics, etc.)
Price	DECIMAL(10,2)	Unit price of the product
Quantity	INT	Number of units ordered
TotalSales	DECIMAL(10,2)	Line total (Price × Quantity)
CustomerName	VARCHAR(100)	Full name of the customer
CustomerLocation	VARCHAR(50)	City of the customer
PaymentMethod	VARCHAR(50)	Payment channel used
Status	VARCHAR(50)	Completed / Pending / Cancelled

Table 3: Complete data dictionary for `amazon_sales`

Design note: Using `DECIMAL(10,2)` for `Price` and `TotalSales` is a deliberate best practice for financial data. Even though the current sample contains whole-dollar amounts, the type protects against future fractional prices and avoids floating-point rounding issues that can arise with `FLOAT` or `DOUBLE`.

7. Data Quality Assessment & Preparation

Before any analytical query was executed, a systematic data quality review was performed. The goals were to confirm completeness, consistency, and readiness for aggregation.

Checks Performed

- **Null scan:** Every column was tested for NULL values. No nulls were found in the final loaded table.
- **Row count validation:** Confirmed exactly 250 rows after load, matching the source CSV.
- **Date range verification:** Minimum and maximum dates confirmed as February 2025 through early April 2025.
- **Consistency check:** Verified that $\text{TotalSales} = \text{Price} \times \text{Quantity}$ for sampled rows (within rounding tolerance).
- **Category and status integrity:** Confirmed only expected values appear in Category, PaymentMethod, and Status fields.
- **Primary key uniqueness:** OrderID is unique and serves as the primary key.

Preparation Steps

1. Created a dedicated MySQL database (amazon_dataset) with a single fact table (amazon_sales).
2. Defined appropriate data types and a primary key constraint.
3. Loaded the cleaned CSV, converting date strings into proper DATE values.
4. Ran exploratory queries (COUNT, null checks, distinct value lists) to validate readiness.
5. Proceeded to the ten business analysis queries only after quality checks passed.

Because the source file was already relatively clean, no extensive transformation or outlier removal was required. The emphasis remained on correct typing, referential integrity within the single table, and reproducible loading scripts.

8. Tools, Environment & Technology Stack

The project uses a deliberately lean but professional technology stack that is widely recognized in industry and accessible for portfolio demonstration.

Layer	Technology	Role in Project
Database	MySQL 8.0+	Storage, schema, advanced SQL analysis
Query Language	SQL (ANSI + MySQL extensions)	CTEs, window functions, aggregations
Visualization	Power BI Desktop	Interactive multi-page dashboard
Source Data	CSV	Initial import into MySQL / Power BI
Documentation	Markdown + PDF + PPTX	README, insights, presentation, this report
Version Control	Git + GitHub	Project packaging and portfolio hosting

Table 4: Technology stack

MySQL was chosen because it is open-source, widely used, and fully supports the advanced SQL features required (window functions, CTEs). Power BI was selected for its strong dashboarding capabilities, ease of sharing via .pbix templates, and industry relevance for business intelligence roles. The combination allows the same cleaned dataset to be queried both programmatically (SQL) and interactively (Power BI).

9. Analytical Methodology

The analytical approach follows a structured sequence that ensures reproducibility and clear linkage between business questions and technical implementation.

Phase 1 — Database Foundation

A dedicated database and fact table were created. Data types and constraints were defined before any load. The CSV was imported and immediately validated with row counts and null checks. This phase establishes a trustworthy analytical base.

Phase 2 — Exploratory Data Analysis (EDA)

Simple descriptive queries were run to understand distributions of Category, Status, PaymentMethod, and monthly volume. These queries informed the design of the ten formal business questions and the layout of the Power BI pages.

Phase 3 — Structured Business Questions

Ten questions were written in plain business language and then translated into SQL. Questions progress from simple retrieval and aggregation to more advanced ranking using window functions. Each question has a clear owner metric and a defined output shape (list, ranked table, percentage, etc.).

Phase 4 — Interpretation & Insight Generation

Query results were examined not only for numerical correctness but for business meaning. Concentration of revenue, cancellation magnitude, and customer concentration were highlighted as the most important signals.

Phase 5 — Visualization & Communication

Findings were mapped onto a four-page Power BI dashboard. KPI cards, ranking charts, maps, and status breakdowns were designed so that a stakeholder can answer the same ten questions interactively. Supporting documentation (this report, README, insights file, and presentation) was produced to complete the delivery package.

10. SQL Business Questions & Solutions

This section documents each of the ten business questions, the analytical intent, the SQL technique employed, and a brief note on interpretation. Full query text is maintained in the project's `sql/` folder.

Question 1 — Completed Orders in March 2025

Business Intent: Retrieve all completed orders placed in March 2025, showing OrderID, Date, Product, and TotalSales, sorted by date descending.

SQL Technique: Date filtering with LIKE '2025-03%' (or BETWEEN), Status filter, ORDER BY. Demonstrates basic retrieval and temporal slicing.

Interpretation Note: Provides a clean list of successful March transactions for operational review.

Question 2 — Overall Revenue, Quantity & AOV

Business Intent: Calculate total revenue, total quantity sold, and average order value across the entire dataset.

SQL Technique: SUM, AVG, ROUND. Single-pass aggregation over the fact table.

Interpretation Note: Establishes the headline numbers used throughout the rest of the analysis and the dashboard KPI cards.

Question 3 — Category Performance

Business Intent: Show total sales and number of orders for each category, sorted by total sales descending.

SQL Technique: GROUP BY Category, SUM, COUNT, ORDER BY. Classic categorical aggregation.

Interpretation Note: Reveals extreme concentration: Electronics and Home Appliances dominate revenue.

Question 4 — Top 5 Products by Quantity

Business Intent: Identify the five best-selling products by total units sold.

SQL Technique: GROUP BY Product, SUM(Quantity), ORDER BY, LIMIT 5.

Interpretation Note: Smartwatch, Smartphone, and Laptop appear at the top, informing inventory and promotion focus.

Question 5 — Top 5 Customers by Revenue

Business Intent: Rank customers by total sales and return the top five with order counts.

SQL Technique: GROUP BY CustomerName, SUM, COUNT, ORDER BY, LIMIT.

Interpretation Note: Highlights a small set of high-value customers suitable for loyalty treatment.

10. SQL Business Questions & Solutions (continued)

Question 6 — Payment Method Analysis

Business Intent: For each payment method show total sales, order count, and percentage of overall revenue.

SQL Technique: GROUP BY plus a scalar subquery for the grand total; percentage calculation with ROUND.

Interpretation Note: PayPal and Credit Card lead; useful for payment-partner and UX decisions.

Question 7 — Status-wise Performance & Completion Rate

Business Intent: Break down orders and sales by Status and compute overall completion rate.

SQL Technique: GROUP BY Status, conditional aggregation with CASE WHEN, percentage of total.

Interpretation Note: Surfaces the 30.8% cancellation rate as the primary operational red flag.

Question 8 — Top 5 Cities by Sales

Business Intent: Rank cities by total sales and also report average order value and order count.

SQL Technique: GROUP BY CustomerLocation, SUM, AVG, COUNT, ORDER BY, LIMIT.

Interpretation Note: Identifies priority markets (Miami, Denver, Houston, Dallas, etc.).

Question 9 — Monthly Sales Trend

Business Intent: Show monthly total sales and order counts from February to April 2025.

SQL Technique: DATE_FORMAT for month bucketing, WHERE date range, GROUP BY month.

Interpretation Note: February is the strongest month; April is thinner (partial month / lower volume).

Question 10 — Top 3 Products per Category (Window Function)

Business Intent: Within each category, rank products by total sales and return only the top three.

SQL Technique: CTE for product-level aggregation, DENSE_RANK() OVER (PARTITION BY Category ORDER BY total_sales DESC), outer filter on rank ≤ 3 .

Interpretation Note: Demonstrates advanced SQL and produces a category-aware product ranking for assortment decisions.

11. Key Findings & Insights

The following findings synthesize the results of the ten SQL questions and the Power BI exploration. They are ordered by business importance.

11.1 Revenue Concentration

Electronics contributes approximately 53% of total revenue and Home Appliances approximately 43%. Together they exceed 96% of all sales. Clothing, Footwear, and Books are marginal from a revenue perspective. This extreme concentration implies that growth strategies and inventory risk management should focus overwhelmingly on the two high-ticket categories.

11.2 Cancellation Rate as Critical Risk

Nearly 31% of all orders are cancelled, representing a large leakage of potential revenue and a likely source of customer dissatisfaction. Pending orders add another 34%, leaving only 35% fully completed. The combination of high cancellation and high pending volume is the single most important operational finding of the project and should be the first area for process investigation.

11.3 Product Leaders

By revenue the leaders are Refrigerator, Laptop, and Smartphone. By quantity the leaders are Smartwatch, Smartphone, and Laptop/Headphones. Smartwatch's high unit volume at a moderate price point makes it an interesting promotional and bundling candidate, while Refrigerator and Laptop drive absolute revenue.

11.4 Customer Concentration

A relatively small number of named customers (Olivia Wilson, Jane Smith, Emma Clark, and others) account for a material share of revenue. This creates both an opportunity (targeted loyalty and upsell) and a risk (over-dependence on a few relationships).

11.5 Geographic Patterns

Sales are concentrated in major US cities. Miami, Denver, Houston, Dallas, Seattle, Boston, Chicago, New York, Los Angeles, and San Francisco appear repeatedly among top locations. Regional marketing and logistics prioritization can be guided by this ranking.

11.6 Payment Method Mix

PayPal and Credit Card together capture the majority of revenue. Gift Card remains meaningful. Debit Card and Amazon Pay trail. Payment experience optimization and fee negotiation can be focused on the leading methods.

11.7 Temporal Pattern

February records the highest sales, followed by March; April is substantially lower (partly because the data window ends early in the month). Any seasonal campaign planning should treat February–March as the stronger window within this sample.

12. Power BI Dashboard Design

The Power BI dashboard translates the SQL findings into an interactive experience. It contains four pages, each with a distinct purpose and a consistent visual language (dark headers, clear KPI cards, ranked charts, and status/color coding).

Page 1 — Overview

Executive summary page. KPI cards display Total Orders, Total Revenue, Average Order Value, Completion Rate, and Cancellation Rate. A donut chart shows category contribution. A line chart presents the monthly sales trend. Status slicers allow quick filtering. This page answers “How are we doing overall?” in a few seconds.

Page 2 — Sales Analysis

Product and category deep-dive. Horizontal bar chart of top products by revenue, donut or pie of category share, dual-axis chart of monthly sales versus order count, and a supporting table of product-level totals. Designed for merchandising and category managers.

Page 3 — Customer Insights

Customer and geographic view. Map visual of sales by city, ranked bar chart of top customers, table of customer metrics (sales, order count, location), and average sales per customer KPI. Supports CRM and regional strategy conversations.

Page 4 — Operations

Operational health page. Revenue by payment method, stacked bar of order status by category, status summary table with percentages, and cancellation rate highlight. Directly addresses the cancellation and pending-order findings.

Slicers for Date, Category, Status, and Payment Method are available across pages so users can explore interactively while remaining within the same analytical frame established by the SQL questions.

13. Business Recommendations

Recommendations are ordered by expected impact and feasibility given the data available.

1. Treat Cancellation Reduction as Priority One

A 30.8% cancellation rate is unsustainable. Initiate a root-cause analysis focusing on Electronics and Home Appliances. Examine inventory accuracy, delivery promises, pricing competitiveness, and post-order communication. Even a modest reduction in cancellations would recover material revenue.

2. Double Down on High-Ticket Categories

Electronics and Home Appliances generate nearly all revenue. Allocate marketing budget, inventory investment, and promotional energy primarily to these categories. Low-ticket categories can remain for assortment completeness but should not drive strategic focus.

3. Build Loyalty Programs for Top Customers

A small set of customers contributes a large share of sales. Design recognition, exclusive offers, or early-access programs for names such as Olivia Wilson, Jane Smith, and Emma Clark to protect and grow their lifetime value.

4. Optimize the Payment Experience

PayPal and Credit Card already lead. Reduce friction for these methods, negotiate favorable rates, and consider light incentives. Monitor Gift Card usage for promotional campaign effectiveness.

5. Prioritize High-Performing Cities

Concentrate regional marketing, faster shipping options, and local partnerships in cities that already show strong sales (Miami, Denver, Houston, Dallas, Seattle, and similar).

6. Clear the Pending Order Backlog

Pending orders represent 34% of volume. Review fulfillment SLAs, stock availability signals, and customer follow-up processes so that a higher share of pending orders convert to completed status.

14. Limitations & Assumptions

Every analysis has boundaries. The following limitations and assumptions should be kept in mind when interpreting results or extending the work.

- **Sample size:** 250 orders is sufficient for technique demonstration and directional insight but is not large enough for robust statistical inference or rare-event modeling.
- **Time window:** Only three months of data are available; seasonality beyond Q1 cannot be assessed.
- **No cost data:** Revenue is analyzed without cost of goods, shipping cost, or payment fees, so profit cannot be calculated.
- **No customer demographics:** Beyond name and city, demographic or behavioral attributes are absent.
- **Status causality unknown:** The data shows cancellation volume but does not record reason codes, so root-cause hypotheses remain qualitative.
- **Single table design:** A pure fact table without separate dimension tables was used for simplicity; a fuller star schema would be preferred at larger scale.
- **Static snapshot:** The analysis is point-in-time; no incremental refresh or streaming architecture is implemented.

These limitations do not invalidate the findings within the given scope; they simply define the safe boundary of interpretation and point to natural next steps for a production analytics environment.

15. Future Enhancements

The current project establishes a solid foundation. The following extensions would increase both analytical depth and operational value.

- **RFM Customer Segmentation:** Score customers on Recency, Frequency, and Monetary value to create actionable segments (Champions, At Risk, etc.).
- **Cancellation Prediction:** Build a classification model using order attributes to flag high-risk orders at placement time.
- **Demand Forecasting:** Apply time-series methods to project monthly or weekly sales for top products.
- **Star Schema & Data Warehouse:** Normalize into fact and dimension tables and move to a proper analytical warehouse.
- **Automated Pipeline:** Schedule CSV or API ingestion with Python or Airflow and refresh the Power BI dataset on a cadence.
- **Real-time Dashboard:** Connect Power BI in DirectQuery mode to a live MySQL or cloud database for near-real-time monitoring.
- **Margin Analysis:** Incorporate cost data to shift the conversation from revenue to contribution margin.
- **Cohort Retention:** Track repeat purchase behavior over time once a longer history is available.

Any of these directions can be pursued as a follow-on portfolio project or as a real initiative inside a business that already owns the transactional data.

16. Skills Demonstrated

PrimeViz was intentionally designed to surface a balanced set of technical and analytical skills that are commonly evaluated for data analyst roles.

Database Design & Modeling: Creation of a clean schema with appropriate data types, primary key, and loading strategy.

Advanced SQL: Proficiency with CTEs, window functions (DENSE_RANK), conditional aggregation, subqueries, date functions, and percentage-of-total calculations.

Exploratory Data Analysis: Systematic null checks, distribution reviews, and consistency validation before formal modeling.

Business Question Framing: Translation of vague business needs into ten precise, answerable questions with clear metrics.

Metric Definition & Governance: Consistent definitions for revenue, AOV, completion rate, and cancellation rate across SQL and Power BI.

Dashboard Design: Multi-page Power BI layout with KPI cards, ranked visuals, maps, and operational views that map directly to the analysis questions.

Data Storytelling: Narrative that moves from observation to interpretation to prioritized recommendation.

Documentation & Reproducibility: Complete README, technical notes, insights document, presentation, and this formal report so that others can understand and extend the work.

End-to-End Ownership: Ability to take a raw CSV through database, analysis, visualization, and stakeholder-ready packaging without external hand-offs.

17. Conclusion

PrimeViz successfully demonstrates a complete, professional analytics workflow. Starting from a raw sales CSV, the project builds a properly typed MySQL database, answers ten substantive business questions with advanced SQL, interprets the results in a business context, and delivers an interactive Power BI dashboard together with comprehensive documentation.

The most important substantive findings are the extreme revenue concentration in Electronics and Home Appliances and the alarmingly high cancellation rate of 30.8%. These two insights alone justify focused action: protect and grow the high-ticket categories, and urgently diagnose and reduce cancellations. Supporting findings on top customers, leading cities, and preferred payment methods provide additional levers for marketing and operations teams.

Beyond the specific numbers, the project illustrates a repeatable method: define clear questions, enforce data quality, choose the right SQL techniques, visualize for decision makers, and document thoroughly. That method is the transferable value of the work and the reason PrimeViz is suitable for a professional data analyst portfolio.

Future iterations can extend the foundation into predictive modeling, customer segmentation, and automated pipelines. The current deliverable already stands as a coherent, end-to-end example of turning raw sales data into actionable intelligence.

Appendix A — Sample SQL Queries

Representative query patterns used in the project are shown below for reference. The complete scripts reside in the `sql/` directory of the repository.

A.1 Overall Metrics

```
SELECT SUM(TotalSales) AS total_revenue, SUM(Quantity) AS total_quantity_sold,  
ROUND(AVG(TotalSales), 2) AS average_order_value FROM amazon_sales;
```

A.2 Category Performance

```
SELECT Category, SUM(TotalSales) AS total_sales, COUNT(*) AS number_of_orders FROM amazon_sales  
GROUP BY Category ORDER BY total_sales DESC;
```

A.3 Payment Method with Revenue Share

```
SELECT PaymentMethod, SUM(TotalSales) AS total_sales, COUNT(*) AS number_of_orders,  
ROUND(SUM(TotalSales)*100.0/(SELECT SUM(TotalSales) FROM amazon_sales), 2) AS revenue_percentage  
FROM amazon_sales GROUP BY PaymentMethod ORDER BY total_sales DESC;
```

A.4 Top 3 Products per Category (Window Function)

```
WITH product_sales AS (SELECT Category, Product, SUM(TotalSales) AS total_sales FROM amazon_sales  
GROUP BY Category, Product), ranked_sales AS (SELECT Category, Product, total_sales, DENSE_RANK()  
OVER (PARTITION BY Category ORDER BY total_sales DESC) AS rank_value FROM product_sales) SELECT  
Category, Product, total_sales, rank_value FROM ranked_sales WHERE rank_value <= 3 ORDER BY  
Category, rank_value;
```

These patterns—simple aggregation, percentage-of-total via subquery, and partitioned ranking—cover the majority of analytical needs encountered in the project and are directly reusable in larger environments.

Appendix B — Project File Structure

The repository is organized for clarity and ease of navigation by recruiters, reviewers, and future maintainers.

README.md — Primary project overview, objectives, insights summary, and usage instructions.

data/amazon_sales_2025.csv — Cleaned source dataset used for both MySQL load and Power BI import.

sql/01_database_setup.sql — Database and table creation script with proper data types.

sql/02_business_analysis.sql — All ten business questions in clean, commented form.

sql/full_analysis.sql — Original combined script including exploration steps.

powerbi/sales2025.pbit — Power BI template containing the four dashboard pages.

powerbi/README.md — Instructions for opening and connecting the template.

docs/insights.md — Concise findings and recommendations document.

docs/technical_notes.md — Schema decisions, SQL techniques, and reproducibility notes.

docs/PrimeViz_Project_Presentation.pptx — 12-slide professional presentation.

docs/PrimeViz_Full_Project_Documentation.pdf — This complete formal report.

.gitignore / .gitattributes — Standard exclusions and line-ending rules for the repository.

This structure separates data, code, visualization, and documentation so that each artifact can be reviewed or reused independently while still forming a coherent whole.

Appendix C — Glossary of Terms

AOV (Average Order Value): Total revenue divided by number of orders; a core e-commerce efficiency metric.

Cancellation Rate: Share of orders with Status = Cancelled; treated as a primary operational risk indicator in this project.

Completion Rate: Share of orders with Status = Completed.

CTE (Common Table Expression): Named temporary result set defined with WITH, used for readable multi-step SQL.

DENSE_RANK: Window function that assigns consecutive ranks without gaps after ties.

Fact Table: Central table in an analytical model containing measurable events (here, orders) and foreign keys or attributes.

KPI: Key Performance Indicator; a headline metric tracked on the dashboard (revenue, rates, etc.).

Partition (SQL): In window functions, the set of rows over which ranking or aggregation is calculated independently.

Power BI Template (.pbit): Portable Power BI file that contains report layout and can be pointed at a data source.

Revenue Share / Percentage of Total: A category or method's sales divided by grand total sales, expressed as a percentage.

Star Schema: Analytical modeling pattern with a central fact table linked to dimension tables (not fully implemented here for simplicity).

Window Function: SQL function that performs calculation across a set of rows related to the current row without collapsing the result set.

End of Document — PrimeViz E-Commerce Sales Analysis Full Project Documentation

Turning raw sales data into actionable intelligence.